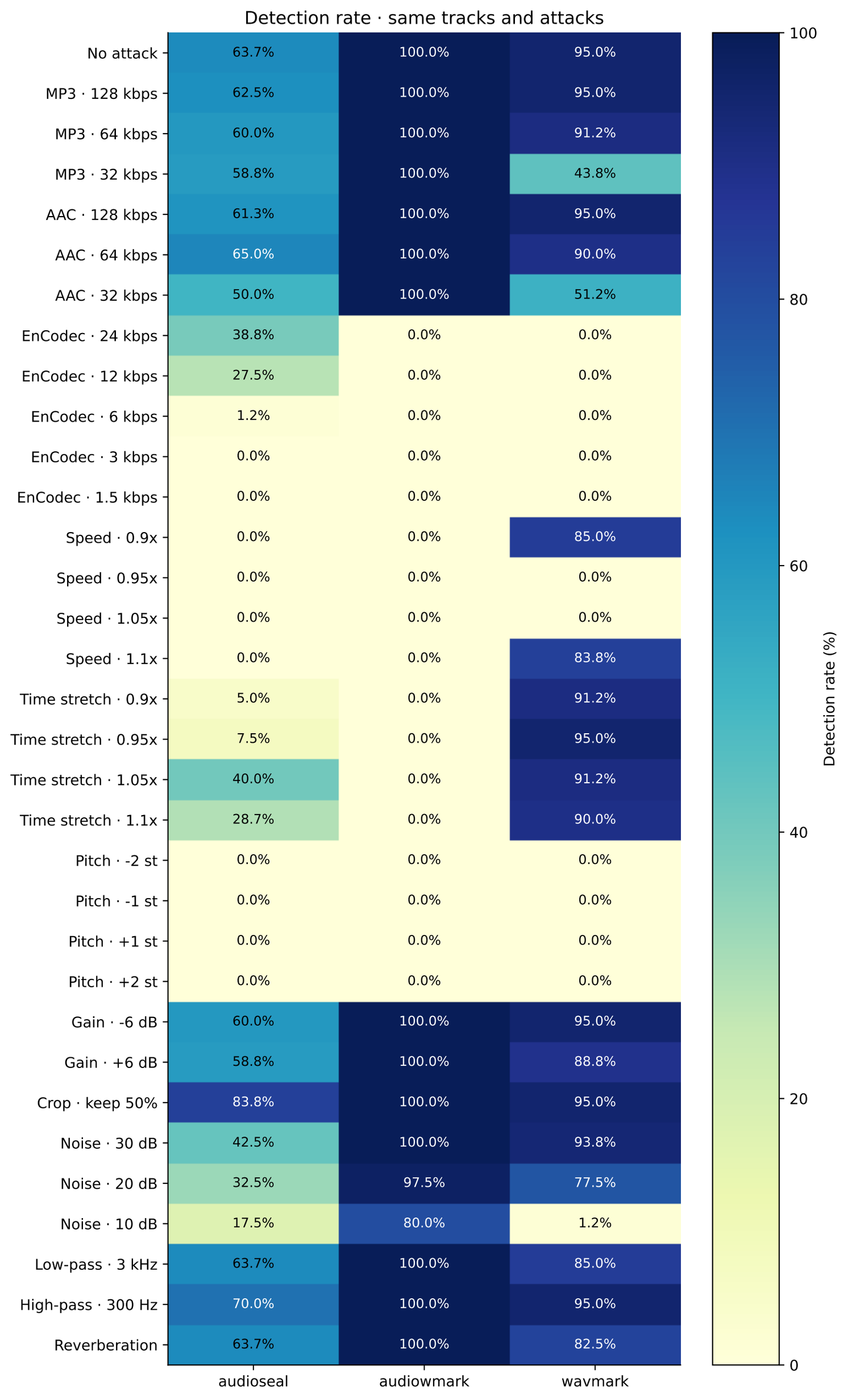
Detection rate · same tracks and attacks
No attack
MP3 · 128 kbps
MP3 · 64 kbps
MP3 · 32 kbps
AAC · 128 kbps
AAC · 64 kbps
AAC · 32 kbps
EnCodec · 24 kbps
EnCodec · 12 kbps
EnCodec · 6 kbps
EnCodec · 3 kbps
EnCodec · 1.5 kbps
Speed · 0.9x
Speed · 0.95x
Speed · 1.05x
Speed · 1.1x
Time stretch · 0.9x
Time stretch · 0.95x
Time stretch · 1.05x
Time stretch · 1.1x
Pitch · -2 st
Pitch · -1 st
Pitch · +1 st
Pitch · +2 st
Gain · -6 dB
Gain · +6 dB
Crop · keep 50%
Noise · 30 dB
Noise · 20 dB
Noise · 10 dB
Low-pass · 3 kHz
High-pass · 300 Hz
Reverberation
63.7% 100.0% 95.0%
62.5% 100.0% 95.0%
60.0% 100.0% 91.2%
58.8% 100.0% 43.8%
61.3% 100.0% 95.0%
65.0% 100.0% 90.0%
50.0% 100.0% 51.2%
38.8% 0.0% 0.0%
27.5% 0.0% 0.0%
1.2% 0.0% 0.0%
0.0% 0.0% 0.0%
0.0% 0.0% 0.0%
0.0% 0.0% 85.0%
0.0% 0.0% 0.0%
0.0% 0.0% 0.0%
0.0% 0.0% 83.8%
5.0% 0.0% 91.2%
7.5% 0.0% 95.0%
40.0% 0.0% 91.2%
28.7% 0.0% 90.0%
0.0% 0.0% 0.0%
0.0% 0.0% 0.0%
0.0% 0.0% 0.0%
0.0% 0.0% 0.0%
60.0% 100.0% 95.0%
58.8% 100.0% 88.8%
83.8% 100.0% 95.0%
42.5% 100.0% 93.8%
32.5% 97.5% 77.5%
17.5% 80.0% 1.2%
63.7% 100.0% 85.0%
70.0% 100.0% 95.0%
63.7% 100.0% 82.5%
audioseal
audiowmark
wavmark
Detection rate (%)
100
80
60
40
20
0